

Section: Particles and Waves

Topic: Radiation, The Inverse Square Law

A radioactive source emits gamma radiation.

A detector is placed at various distances and the corrected count rate is recorded:

Distance from source (m)	Count rate (counts/s)
0.20	400
0.40	100
0.60	44
0.80	25

(a)

Explain why the source must be small compared to the distances used.

✔ So that the source acts as a **point source**, ensuring that the gamma radiation spreads out unif

If the source were large, the inverse square law would not apply accurately because the distanc
source.

(b)

Show that the results are consistent with the inverse square law.

The **inverse square law** states:

$I \propto \frac{1}{d^2}$

So, calculate $\text{Count rate} \times d^2$ for each data point:

- At 0.20 m: $400 \times 0.20^2 = 400 \times 0.04 = 16$
- At 0.40 m: $100 \times 0.40^2 = 100 \times 0.16 = 16$
- At 0.60 m: $44 \times 0.60^2 = 44 \times 0.36 = 15.84$
- At 0.80 m: $25 \times 0.80^2 = 25 \times 0.64 = 16$

✔ All values are approximately constant → confirms that:

$I \propto \frac{1}{d^2}$

🧠 Revision Tips:

- The inverse square law applies to **point sources** spreading out in **three dimensions**
- Always check $I \times d^2$ is constant
- Units don't matter — you're checking **proportionality**, not absolute values